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Inventor(s)	Merson; Alexander et al.

Syringe control structure and manufacture method

Abstract

An improved hypodermic syringe including finger-holding surfaces that substantially or completely surround the operator finger and include an interior tapered portion to better engage a finger, which facilitates one-handed use by allowing the same hand that holds the syringe to operate the plunger conveniently to accomplish both the distal (injecting) stroke and the proximal (aspirating) stroke.

Inventors:	Merson; Alexander (Brooklyn, NY), Kleyman; Gennady I (Brooklyn, NY)
Applicant:	Merson; Alexander (Brooklyn, NY); Kleyman; Gennady I (Brooklyn, NY)
Family ID:	83547079
Assignee:	Algen & Klemer, LLC (New York, NY)
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Primary Examiner: Ponton; James D

Assistant Examiner: Trinh; Hong-Van N

Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to hypodermic syringes structured to facilitate use, in particular, hypodermic syringes having finger receptacles to better engage operator fingers to facilitate single-hand use of the hypodermic syringe.

BACKGROUND OF THE INVENTION

(2) Hypodermic used requires skilled and confident, accurate operation (e.g. a controlled compression of a syringe plunger) and simultaneous precise control of the needle location and depth. Moreover, it is very important to precisely accomplish both the distal (injecting) stroke, e.g. to determine whether the needle is in a blood vessel, and the proximal (aspirating) stroke. Existing syringes typically have limited plunger and barrel body surfaces used to grasp, operate and control dispensing of the syringe content, which syringe surfaces typically minimally accommodate the diverse finger sizes, or have any structure to enhance the operation and control of the syringe by the user.

SUMMARY OF THE INVENTION

(3) Disclosed is an ergonomically improved hypodermic syringe including finger-holding surfaces that substantially or completely surround the operator finger and include an interior tapered portion to better engage a finger, which facilitates one-handed use by allowing the same hand that holds the syringe to operate the plunger conveniently to accomplish both the distal (injecting) stroke and the proximal (aspirating) stroke.

(4) One embodiment includes a ring grip on the operating (finger) end of the syringe plunger, sized and shaped to accommodate or adjust for a snug fit with fingers and thumbs of varying sizes and additional finger holding structure located on the syringe barrel that can accommodate or adjust for a snug fit with index and middle fingers. Another embodiment includes structures that adjust to reduce the space between the grip and better engage the operator finger. Other embodiments apply similar structural improvement to finger grip(s) on the syringe barrel, and separate similarly shaped

finger grips that may be added or attached to an existing syringe plunger and barrel.

(5) These embodiments make the syringe significantly more easy, predictable, and controllable to use with one hand. The result is a syringe and syringe attachment(s) which enables a reliable and controlled aspiration maneuver, to test for the presence of blood indicating that a blood vessel has been penetrated, and other critical operations to be performed easily with one hand.

Description

BRIEF DESCRIPTION OF THE DRAWING

(1) These and further novel features of the present invention are better understood by taking the following Drawing figures together with the Detailed Description, wherein:

(2) FIG. 1—is an isometric view of the syringe assembly of a first exemplary embodiment of the present invention;

(3) FIG. 2—is a front elevation view of the syringe assembly by the embodiment of FIG. 1;

(4) FIG. 3—is a section 3-3 of the embodiment of FIG. 2;

(5) FIG. 4—is a section 4-4 of the embodiment of FIG. 2;

(6) FIG. 5—is an isometric view of a syringe assembly by a second exemplary embodiment of the present invention;

(7) FIG. 6—is a front elevation view of the syringe assembly by the embodiment of FIG. 2;

(8) FIG. 7—is a section 7-7 of the embodiment of FIG. 6;

(9) FIG. 8—is a section 8-8 of the embodiment of FIG. 6;

(10) FIG. 9—is an exploded view of syringe a barrel and finger holding attachment embodiment;

(11) FIG. 10—shows an isometric view of a syringe barrel and finger holding attachment assembly;

(12) FIG. 11—shows sectional view 11-11 of the embodiment of FIG. 6;

(13) FIG. 12, FIG. 13 and FIG. 14—shows a top, and side and front elevation views of barrel finger holding attachment of the embodiment of FIG. 9;

(14) FIG. 15—shows an isometric view of a syringe assembly by a third exemplary embodiment of the present invention;

(15) FIG. 16—shows a front elevation view of the syringe assembly by of the embodiment of FIG. 15;

(16) FIG. 17—shows a section 17-17 of the embodiment of FIG. 10;

(17) FIG. 18—shows a section 18-18 of the embodiment of FIG. 10;

(18) FIG. 19—shows an exploded isometric view of a syringe assembly by a fourth exemplary embodiment of the present invention;

(19) FIG. 20 and FIG. 21—are front elevation view of the syringe assembly by the embodiment of FIG. 19 of the present invention showing different screw 178 positions;

(20) FIG. 22—shows an isometric view of a syringe assembly by a fifth exemplary embodiment of the present invention;

(21) FIG. 23—shows an exploded isometric view of the syringe assembly by a the embodiment of FIG. 22;

(22) FIG. 24—shows a front elevation view of the syringe assembly of the embodiment of FIG. 22;

(23) FIG. 25—shows a side elevation view of the syringe assembly of the embodiment of FIG. 22, when the upper ring portion is in a top position

(24) FIG. 26—shows a section 26-26 of the embodiment of FIG. 25;

(25) FIG. 27—is an enlarged view 27 of the embodiment of FIG. 26;

(26) FIG. 28—shows a partially side view of the syringe assembly of the embodiment of FIG. 22, when the upper ring portion is in a bottom position;

(27) FIG. 29—shows a section 29-29 of the embodiment of FIG. 28;

(28) FIG. 30—is an enlarged view 30-30 of the embodiment of FIG. 29;

- (29) FIG. 31—is isometric view of the plunger with ring bottom portion of the embodiment of FIG. 22;
- (30) FIG. 32—is an enlarged view 32 of the embodiment of FIG. 31;
- (31) FIG. 33—is an isometric view of the ring upper portion of the embodiment of FIG. 22;
- (32) FIG. 34—is a side elevation view of ring upper portion of the embodiment of FIG. 22;
- (33) FIG. 35—is a section 35-35 of the embodiment of FIG. 34;
- (34) FIG. 36—is an enlarged view 36 of the embodiment of FIG. 35;
- (35) FIG. 37—shows an isometric view of a syringe assembly by a sixth exemplary embodiment of the present invention;
- (36) FIG. 38—shows an exploded isometric view of syringe assembly by the embodiment of FIG. 37;
- (37) FIG. 39—shows an isometric view of an attachable ring embodiment of the present invention;
- (38) FIG. 40—is a front elevation view of the attachable ring of the embodiment of FIG. 39;
- (39) FIG. 41—is a bottom view of the attachable ring of the embodiment of FIG. 39;
- (40) FIG. 42—is an isometric view of syringe assembly by a seventh exemplary embodiment of the present invention;
- (41) FIG. 43 and FIG. 44—is a front and side elevation view respectively, of syringe assembly by embodiment of FIG. 42;
- (42) FIG. 45—is a section 45-45 of the embodiment of FIG. 43;
- (43) FIG. 46—is an enlarged view 46 of the embodiment of FIG. 45;
- (44) FIG. 47—is an isometric view of syringe assembly of the embodiment of FIG. 42;
- (45) FIG. 48 and FIG. 49—are front and side elevation views of syringe assembly of the embodiment of FIG. 42;
- (46) FIG. 50—is a section 50-50 of the embodiment of FIG. 48;
- (47) FIG. 51—is an enlarged view 51 of the embodiment of FIG. 50;
- (48) FIG. 52, FIGS. 53 and 54—are front and side elevation, and top views respectively, of the plunger of the embodiment of FIG. 42; and
- (49) FIG. 55—is an enlarged view 55 of the embodiment of FIG. 53.

DETAILED DESCRIPTION

(50) The embodiments according to the present invention include a generally tubular member (e.g. **110A-110C**, **110F-110G**) that substantially surround a finger or fingers inserted therein, and further include a tapered surface or surfaces (e.g. **122**, **124**, **140**, **160**, **222**, **224**) that diminish the cross section opening of the tubular member as you move along the length of the tubular member, or adjustable structures (e.g. **110D**, **110E**) that comprises structures (e.g. **178** and **190**, **192**) that adjust to more closely grip a finger inserted within the structures.

(51) As applied to a first exemplary embodiment **60** of FIGS. 1-4, a plunger includes a plunger rod **106** ring structure **110A** that is a finger receiver with a bottom portion **130** attached to plunger rod **106** end, and has a tapered upper and sides **122**, **124** surfaces, wherein the tapered surfaces of the ring **110A** accommodate, snag or otherwise engage different sizes of fingers. The syringe barrel **100** includes finger grips **104A** that have a sloped surface proximal to said plunger rod **106** facing an opposite finger grip, with diminishing internal clearance there between (e.g. moving from distance D1 to D2 shown in FIG. 2) for distances closer to the syringe barrel **100**. In this embodiment, the taper of the proximal surface is mildly arcuate, as shown in FIGS. 1, 2 and in later figures. The needle **102** is connected to the barrel **100** at an end distal to the end from which the plunger rod **106** emerges, and the needle **102** receives the material within the barrel **100** as the plunger rod **106** is advanced into the barrel **100**.

(52) As shown on cross-sections FIG. 3 and FIG. 4, the upper portion **122** of the ring structure **110A** includes a tapered inside the ring surface of **122** and **124**, and this taper at an angle **156** is selected to allow internal dimensions “H1” and “H2” in which to receive a finger. Dimension “H1” is larger than dimension “H2” and each taper may extend over the length of the ring **110A**, and this

allowed fingers with different size to be inserted into the ring, and the inward-facing ring tapered surface will accommodate to the finger size in vertical direction. This will restrict finger movement inside the ring in vertical direction.

(53) Additionally, the ring by this embodiment includes a tapered surface **124** in horizontal direction (optionally separated by a recess **126** from the tapered surface **122**) and these tapers provide the opening **120** inner dimensions “H3” and “H4”. This horizontal taper will accommodate the finger horizontal size.

(54) According to human hand scale (see link

<https://ultraxeondotcom.wordpress.com/2016/02/25/human-scale-from-henry-dreyfuss>), thumb finger thickness (vertical dimensions) minimum dimensions for men are typically 17 mm and for women 15 mm with a typical maximum dimensions for men 24 mm and for women 19 mm.

(55) In the application drawings maximum vertical dimensions (finger entrance into the ring) represented by H1 (FIG. 1), H5 (FIG. 7), H7 (FIG. 17), H9 (FIG. 20), H11 (FIG. 26) and H14 (FIG. 50) and preferably this dimension should be slightly larger than 24 mm (typical maximum dimension for men). In the application drawings minimum vertical dimensions represented by H2 (FIG. 3), H6 (FIG. 7), H8 (FIG. 17), H10 (FIG. 21), H12 (FIG. 29) and H13 (FIG. 45) and preferably these dimensions should be slightly larger than 15 mm (minimum dimension for women). This combination of ring (e.g. **122**, etc.), inside vertical taper (e.g. **124**, etc.) dimensions, allowed adapt (snag) to the majority thumb finger size to comfortably perform the proximal (aspirating) stroke.

(56) As result those tapered surfaces inside ring structure will restrict finger motions inside the ring **110A**, allowing controlled injecting and aspiration stroke without having a gap (free finger movement) inside of the ring, because plunger will follow finger movement simultaneously.

(57) In another novel feature of the embodiment **60** having the syringe barrel **100** that includes an open structure finger grip **104A**, that receives an index and middle finger for holding a barrel typically on to two portions on either side of the barrel **100**. To minimize a vertical movement of those fingers during an aspiration (withdrawal) stroke without having a gap this barrel holding structure has an inside tapered shape indicated by dimensions “D1” and “D2” (FIG. 2) and “D3” and “D4” (FIG. 11). According to human hand scale (see above link), maximum index and middle finger thickness for men is typically 19 mm and minimum typically is 13 mm, and the typical maximum index and middle finger thickness for women is 16 mm and minimum is 11 mm, and the maximum width (horizontal dimensions) for men's index finger is 21 mm and for women is 18 mm and smallest width accordantly for men 15 mm and for women 13 mm. For middle finger for men the maximum is 21 mm and for women 18 mm and minimum width for men 15 mm and for women 13 mm. In most cases barrel finger grips (**104A** and **104B**) will hold by the index and middle fingers.

(58) According to this data, in the embodiments of FIGS. 1-4, the finger entrance **120** into the barrel finger holder structure (dimensions “D1” and “D3”) should be slightly larger than 19 mm and smallest dimensions (“D2” and “D4”) should be around 11 mm and the open loop finger grip (**104A**) dimension “L1” see FIG. 2 to have reliable finger gripping (or ‘snag’) should be more than half of maximum index and middle fingers width. In this case it should be more than 11 mm (21 mm max). By entering a finger within spacing D1, the substantially planar member **105A** attached to and extending away from the barrel and closer to the needle **102** end of the barrel **100** in this exemplary embodiment together with the facing member **105B** also attached to and extending away from the barrel, that includes a mildly curved taper (to more completely engage a somewhat oval finger) providing a diminishing spacing relative the planar member **105A** that grips a finger inserted between members **105A** and **105B** as a finger (or fingers on opposing sides of the barrel **100**), provide increased force on the inserted finger and corresponding increased gripping of the barrel **100**. This taper design adapts to (snags) the majority of index and middle fingers size to comfortably perform the retracting (aspirating) stroke.

(59) In case of close loop finger grips (**104B**, discussed below) dimension “L2” of FIG. 6, should be larger than maximum width of index and middle fingers, to allow finger to be inserted into the grip. In this case it should be larger than 21 mm.

(60) In another novel feature of the further embodiment **62** having the syringe barrel **100** that includes, or receives an optionally removable from, a closed short squared tubular finger grip **104B**, that separately receives an index and middle finger for holding the barrel **100**. In the embodiments **62** of FIGS. 5 and 6, finger grip **104B** is shown seated on, or part of the barrel **100**, while the embodiments of FIGS. 9, 10 show the finger grip **104B** separate from the barrel **100** but is capable of being seated on the barrel shoulder or collar **101** within a recess **164** (FIG. 14) formed by a ring segment **162** that surrounds the barrel **100** and forms the recess **164** relative to the top of the finger grip **104B**.

(61) The finger grip **104B** includes two finger grip sections disposed on opposite sides of the barrel **100**, and each section includes an internal surface **160** inclined at an angle **156** relative to the top **154** of the finger grip to provide an internal opening that tapers from an opening dimension D3 to D4 along the width of the of the finger grip sections. As the plunger is advanced into the barrel **100**, the operator finger engages (applies force) on to the surfaces **160**. The finger grip **104B** bottom of **152** is disposed to provide openings according to the criteria described above in regard to the embodiments of FIGS. 1-4. The top **154** of the finger grip **104B** includes an opening **158** though the plunger extends when the finger grip **104B** is seated on the barrel **100**. The dimensions D3, D4 of FIG. 11 are equivalent to the dimensions H3, H4 shown in FIG. 3.

(62) A further embodiment **64** is shown in FIGS. 15-18, wherein the plunger **100** finger grip ring **110C** includes a slightly curved upper portion **144** attached to a bottom portion **138** by sides **142** typically having uniform thickness (but adaptable to be tapered as shown in FIG. 4 in other embodiments), wherein the bottom portion **144** includes a tapered surface **140**, analogous to the tapered portion **122** of FIGS. 1-7, but applied to the ring **110C** proximal to the plunger rod **106**. The opening heights H7 and H8 correspond to finger entrance and exit of other embodiments herein.

(63) A fourth embodiment **66** is shown in FIGS. 19-21, wherein the plunger **106** ring **110D** comprises a closed annular structure having substantially (but not necessarily) uniform side wall thickness including walls **172** proximal to the sides of fingers inserted therein, a slightly (but not necessarily) curved bottom **170** and curved top **174** having a spacing H9 and a threaded aperture **176** in the curved top **174** that receives a corresponding threaded screw **178** that may be advance (e.g. distance H10) into the ring **110D** to secure or limit movement of a finger inserted therein. Finger grips **104A**, attached to the barrel **100**, are also included in this embodiment.

(64) A fifth embodiment **68** is shown in FIGS. 22-36 comprising a two-part ring **110E** having a wall including a bottom part or curved lower portion **109** attached to the plunger rod **106** and having at least one toothed surface **196**, and an upper portion having a curved upper portion **192** which overlays and/or engages the curved lower portion **190** and has at least a toothed finger **194** having a pawl tooth **204** that engages the toothed surface **196**, each combination of at least the pawl tooth **204** and the toothed surface **196** together form each of two opposing wall segments of the ring **110E** wall are adjustable in the direction of said axis of motion thus providing a selectively diminished opening space in the direction of the axis of the plunger rod **106** while maintaining a substantially uniform dimensions across the opening space between the two opposing segments and perpendicular to said axis of motion, that together work to allow the ring **110E** upper and lower portions to be moved together to decrease the space (that receives a finger) by compression pressure on the ring **110E** and restrict increasing the space from a ring **110E** upper and lower part separating force. Finger grips **104A**, attached to the barrel **100**, are also included in this embodiment.

(65) A vertical cross-section 26 of the embodiment **68** shown in FIG. 25 is shown in FIG. 26, wherein the maximum vertical finger opening H11 between curved upper and lower portions **190** and **192** respectively, is shown with the portions engaged and secured by toothed surface **196** and

pawl tooth **204**, shown in enlarged detail in FIG. 27. A minimum finger distance H12 is shown in FIG. 29 as the cross-section of FIG. 28, wherein the curved upper and curved lower portions **192**, **190** are substantially completely engaged, and the engaged detail is shown in FIG. 30 wherein the pawl **194** tooth **204** engages the end of the toothed surface proximal the bottom of curved lower portion **190**.

(66) Similarly, smooth surfaces **206** adjacent the toothed surface **196** on curved lower portion **190** are shown in the enlarged portion FIG. 32 of FIG. 31, which receive smooth surfaces of members **208** of curved upper portion **192** as shown in FIGS. 33, 34. The finger **194** is separated from and parallel to smooth surface members **208** to allow pawl tooth **204** to engage the complementarily toothed surface **196** without being restricted by the smooth surface members **208**. Smooth surface members **208** may also comprise an “L” shaped cross-section to engage the edges of the curved lower portion **190**. A further view of only the curved upper portion **192** of ring **110E** is shown in FIG. 35 and detail FIG. 36, wherein the engaging point of the pawl tooth **204** is substantially at the end of the side of the “L” shaped smooth surface member **208**.

(67) A sixth embodiment **70** is shown in FIGS. 37-41 having a removable plunger finger ring **110F** which engages and is retained on a standard plunger rod **106A** or custom plunger. As shown in more detail in FIGS. 39-41, the ring **110F** typically (but not necessarily) comprises a continuous and substantially uniform wall thickness ring having top **214**, sides **212**, and bottom **210**, wherein the bottom includes inward-extending locating protrusions **244** that laterally engage the plunger rod **106A** handle **107** when the handle is seated within aperture **211** between prongs **232** (or other retaining devices) to reduce relative motion therebetween upon application of force to the ring **110F** by a finger. To permit resilient grasping of the plunger rod **106A**, the prongs **232** are partially separated by spaces **236** from the bottom **210**. The side walls **212** and top **214** may also include tapered surfaces **222**, **224**, with or without spaces **226**, analogous to the tapered surfaces and walls of other embodiments shown herein.

(68) A seventh embodiment **72** is shown in FIGS. 42-55, wherein an isometric view of syringe assembly is shown in FIG. 42, front and side elevation views of FIGS. 43 and 44, and includes the barrel **100** and attached needle, with plunger rod **106** and finger grips **104A** (or **104B**), and thumb/finger ring **110G** includes a finger plate **256** attached by a hinge **258** and toothed latch (FIG. 46) that is movable to secure the finger plate **256** into position to establish a minimum distance “H13” (as shown on FIG. 45) to secure a “small” thumb finger by moving the finger holding plate downward to an angle **270** (FIG. 46). An isometric view of syringe assembly by the seventh embodiment initially allows thumb finger holding plate in position to establish an open internal opening distance “H14” (as shown on FIG. 50) to secure a “larger” thumb finger and is adjustable to an even larger opening as desired.

(69) In this embodiment as shown in FIGS. 42-46, the thumb/finger plate **256** includes an arcuate member **262** having a radius center substantially at the location of the hinge **258**, and includes sawtooth-shaped teeth **264** on a surface facing the hinge **258**, and engaging a narrowed edge **266** of the top portion **254** of the thumb/finger ring **110G**. A bottom **250** is connected to the end of the plunger rod **106**, and is also connected to the top portion **254** wall by opposing side walls **252**. To reduce the internal space, a downward (e.g. toward the needle **102**) force or force generally perpendicular to the plane of the top portion **254** narrowed edge is applied allowing the incline of the tooth to slip past the narrowed edge **266** and latch the finger/plate **256** into the next reduced internal space position to allow reducing the internal space, e.g. H13.

(70) The isometric, and front and side elevation views of FIGS. 47-49 illustrate raising or retracting the finger plate **256** to increase the ring **110G** internal opening, wherein the arcuate member **262** having teeth **266** of the arcuate member **262** is moved away from engaging the edge **266** by a force **F** (FIG. 44) on the arcuate member **262** free end generally away from the hinge **258** and a compressing force (upward in the views of FIGS. 47-49) is applied to allow increasing the internal space, e.g. H14. An enlarged view is shown in FIG. 50 showing the engaging of the narrowed edge

266 of the ring 110G top portion 254 engaging a middle range of the arcuate member 262 teeth 264. A further enlargement is shown in FIG. 51 including angle 272 that is less than angle 270 of FIG. 46.

(71) The front, side elevation, and top plan views of plunger rod 106 with thumb/finger ring 110G of this embodiment is shown in FIGS. 52-54 as a single piece fabricated by injection molding. In this embodiment, the plunger rod 106, thumb/finger ring 110G, 'live' (bendable from mold material) hinge 258 attached to thumb/finger ring 110G, finger plate 256, and arcuate member 264 comprise a single mold material or molded structure. After molding and release from the mold (not shown), the finger plate 256 and arcuate member 262 is directed through the ring 110F interior opening, and allow teeth 264 to engage the narrowed edge 266 on top portion 253 at the desired angle, e.g. 270, 272, or otherwise.

(72) Modification and substitutions by one of ordinary skill in the art are within the scope of the present invention, which is not limited except by the claims that follow.

Claims

1. A syringe having improved control members, the syringe comprising: a barrel having a region therein to retain material, and a first opening at one end through which said material is dispensed, and a second opening distal from said first opening; a plunger including a plunger rod disposed to enter said barrel through said second opening and extending into said barrel, and having a first end within said barrel and a second end external to said barrel, and having an axis of motion within said barrel; and a finger receiver connected to said second end, and comprising an annular member having a wall defining an adjustable opening having an opening length extending in a dimension substantially perpendicular to said axis of motion, wherein said annular member wall comprises a lower portion, and an upper portion detached from said lower portion and having a first end and a second end, said upper portion and said lower portion movable together to provide said adjustable opening in a direction of said axis of motion while maintaining a substantially uniform dimension across said adjustable opening and perpendicular to said axis of motion, wherein said first end and said second end of said upper portion each comprises a corresponding first member, and said lower portion comprises a second member adapted to be adjustably received by said first members of said first end and said second end of said upper portion to form said adjustable opening having a wall length adjustable at two locations therealong.
2. The syringe of claim 1, wherein said first members of said first end and second end of said upper portion receive said second member at a region of said second member including a toothed region, wherein said first member of said first end includes a pawl disposed to engage said second member toothed region to retain said first member of said first end of said upper portion at a selected position along said toothed region.
3. The syringe of claim 2, wherein said second member is connected to said plunger and includes a first end and a second end and each of said first end and said second end of said second member having said toothed region thereon, and said first members of said first end and said second end of said upper portion include an "L"-shaped cross-section to engage at least one side of said first end and said second end of said second member.
4. The syringe of claim 2, wherein said upper portion is formed in a "U" shape capable of receiving a finger, said lower portion is formed in a "U" shape capable of receiving a finger and connected to said upper portion to form said adjustable opening, said lower portion being connected to said plunger rod and capable of being displaced by a force directed into said barrel as applied by a finger, and wherein said upper portion and said lower portion engage by a ratchet and pawl connection disposed to prevent displacement of said upper portion relative to said lower portion in a direction opposite of said barrel.
5. A syringe having improved control members, comprising: a barrel comprising a region of rigid

plastic to retain material therein, and a plunger slidably mounted in said barrel having an end extending from said barrel; a plunger finger receiver having an adjustable opening capable of receiving different sizes of fingers, comprising a first end formed in a “U” shape capable of receiving a finger, a second end formed in a “U” shape capable of receiving said finger and connected to said first end to form said opening, said second end connected to said plunger end and capable of being displaced by a force directed into said barrel as applied by a finger, and wherein said first end and said second end engage by a ratchet and pawl connection disposed to prevent displacement of said first end relative to said second end in a direction opposite of said barrel.
